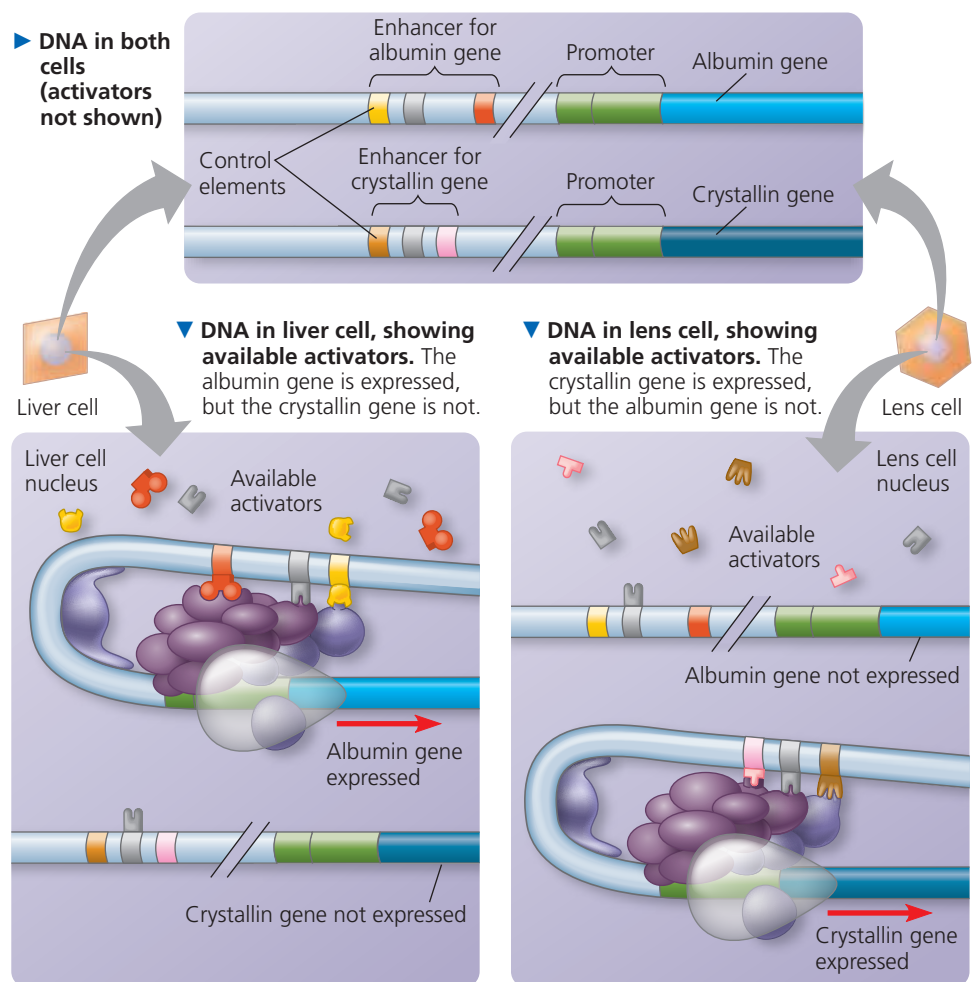


► **Figure 18.12 Cell type-specific transcription.** Both liver cells and lens cells have the genes for making the proteins albumin and crystallin, but only liver cells make albumin (a blood protein) and only lens cells make crystallin (the main protein of the lens of the eye). The specific transcription factors made in a cell determine which genes are expressed. In this example, the genes for albumin and crystallin are shown at the top, each with an enhancer made up of three different control elements. Although the enhancers for the two genes both have a gray control element, each enhancer has a unique *combination* of elements. All the activator proteins required for high-level expression of the albumin gene are present in liver cells only (left), whereas the activators needed for expression of the crystallin gene are present in lens cells only (right). For simplicity, we consider only the role of specific transcription factors that are activators here, although repressors may also influence transcription in certain cell types.

**VISUAL SKILLS** Describe the enhancer for the albumin gene in each type of cell. How would the nucleotide sequence of this enhancer in the liver cell compare with that in the lens cell?



## Nuclear Architecture and Gene Expression

Each chromosome in the interphase nucleus of animal cells occupies a distinct territory (see Figure 16.24a). Recently, *chromosome conformation capture* techniques (3C and others) have been developed that allow researchers to cross-link and identify regions of chromosomes associating with each other during interphase. These studies reveal two organizational details: First, the territory of each chromosome is divided into regions of chromatin loops (*topologically associated domains*, or TADs), within which chromatin sites associate mainly with each other. Second, loops of chromatin, each likely a TAD, extend from individual chromosomal territories into specific sites in the nucleus (**Figure 18.13**). Different loops from the same chromosome and loops from other chromosomes may congregate in such sites, some of which are rich in RNA polymerases and other transcription-associated proteins. Like a recreation center that draws members from many different neighborhoods, these so-called *transcription factories* are thought to be areas specialized for a common function.

The old view that the nuclear contents are like a bowl of amorphous chromosomal spaghetti has given way to a new model of a nucleus with a defined architecture and regulated

movements of chromatin. Several lines of evidence suggest that unexpressed genes are located in the outer edges of the nucleus, while those that are being expressed are found in its interior region. Relocation of particular genes from their chromosomal territories to transcription factories in the interior may be part of the process of readying genes for transcription. In 2017, a consortium of researchers funded by the National Institutes of Health began investigating genome organization over time and its relationship to genome function.

## Mechanisms of Post-transcriptional Regulation

Transcription alone does not constitute gene expression. The expression of a protein-coding gene is ultimately measured by the amount of functional protein a cell makes, and much happens between the synthesis of the RNA transcript and the activity of the protein in the cell. Many regulatory mechanisms operate at the various stages after transcription (see Figure 18.6). These mechanisms allow a cell to rapidly fine-tune gene expression in response to environmental changes without altering its transcription patterns. Here we explore how cells regulate gene expression after transcription.